

Radar HRRP target recognition in frequency domain based on autoregressive model

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Abstract—In this paper, we adopt the autoregressive (AR) model to characterize the frequency spectrum amplitude of high-resolution range profile (HRRP) and extract the AR and partial correlation (PARCOR) coefficients, which are invariant to the initial-phase, translation and scale changes of HRRP, as discriminating features. Moreover, a mixture model based frame partition method is proposed and a Bayesian Ying-Yang (BYY) harmony learning algorithm is adopted to determine the frame number automatically during parameter learning. Experimental results based on measured data demonstrate the proposed features are superior to others in their minor frame number, robustness to sample size and good rejection ability.

I. INTRODUCTION

Radar automatic target recognition (RATR) is to identify the unknown target from its radar echoes. A high-resolution range profile (HRRP) is the coherent summations of the complex time return from target scatterers in each range cell. Since it contains abundant target structure information, radar HRRP target recognition has received intensive attention from the RATR community^[1-7].

Due to the four well-known sensitivity problems, i.e., initial-phase, translation, scale and target aspect sensitivities^[1], the measured HRRP data cannot be directly used for recognition. To cope with these sensitivities, various pre-processing techniques, such as translation alignment, scale normalization and frame partition^[1, 2], have been studied and applied. These pre-processings inevitably increase the system complexity and may bring errors due to algorithm accuracy.

Feature extraction is important in the recognition system. Extracting proper feature not only can reduce system complexity and processing time, but can also improve system performance in many aspects. The works on feature extraction from HRRP concentrate on sensitivity elimination or dimensionality reduction^[3-6]. A lot of features, such as spectra, moments and scatterers' information, have been studied and applied. However, few features available can possess low dimensionality and many invariance properties simultaneously while providing desirable recognition performance.

Wide Sense Stationary-Uncorrelated Scattering model^[8] is widely used in radar signal modeling. According to this model, it can be deduced that the frequency spectrum amplitude of the complex HRRP is stationary. We are thus motivated to employ autoregressive (AR) model to describe the frequency

spectrum amplitude and extract AR coefficients and partial correlation (PARCOR) coefficients as discriminating features. Both of the two features have the following advantages: 1. low feature dimensionality, avoiding curse of dimensionality; 2. translation and scale invariant, simplifying the recognition procedure; 3. sufficient interclass separability; 4. small storage requirement; 5. fast on-line recognition speed;

To deal with the remaining target aspect sensitivity, we still follow a frame partition methodology^[1, 2]. Since the conventional equal interval partition approaches may allocate samples from different distributions into a frame, here we directly adopt a mixture model based frame partition method with a Bayesian Ying-Yang (BYY) learning algorithm to automatically determine the frame number during parameter learning. This method is superior to the traditional two-phase implementation, which suffers from extensive computation and unreliable evaluation to model selection criterion.

This paper is organized as follows. Section 2 analyzes the statistical characteristics of HRRP and introduces the AR model to fulfill the feature extraction task. In section 3, we present a mixture model based frame partition method which is implemented by a BYY learning algorithm. In section 4, several experiments based on measured data are performed to evaluate the recognition performance of the proposed features. Finally, some concluding remarks are drawn in section 5.

II. FEATURE EXTRACTION

A. Statistical Characteristics of HRRP

An HRRP $\mathbf{x}$ is the coherent summations of the complex returns from target scatterers in each range cell. We give the definition of $\mathbf{x}$ and its frequency spectrum $\mathbf{y}$ as follows:

$$\mathbf{x} = e^{j\varphi} [x_1, x_2, x_3, \dots, x_d]^T, \quad i = 1, 2, \dots, d$$
$$\mathbf{y} = \text{fft}(\mathbf{x}) = e^{j\varphi} [y_1, y_2, \dots, y_d]^T, \quad \text{with each } y_f = \sum_{i=1}^d x_i e^{-j2\pi f i / d} \quad (1)$$

where φ is initial phase and f is the index of frequency bin. We try to exploit the frequency spectrum of complex HRRP for target recognition due to its following appealing qualities: 1. FFT is an information preserving procedure, so there is no information lost in the transformation process; 2. Frequency spectrum is translation invariant, so the feature extracted from it can still enjoy this property; 3. A signal represented in the frequency domain may show characteristics which are not normally observed in the time domain (such as stationarity).

According to the Wide Sense Stationary-Unrelated Scattering model^[4], the returns in each range cell are characterized as independent zero-mean complex Gaussian variables. After some mathematical derivation, it is concluded that the correlation function of two frequencies depends only on their difference, i.e., the frequency spectrum is a wide sense stationary process. From (1) we can see the frequency spectrum still suffers from initial-phase sensitivity. So we only consider the amplitude of frequency spectrum $\mathbf{z} = |\mathbf{y}|$ here. It is simple to prove $\mathbf{z}$ is still a stationary series. In the following, we will employ autoregressive model to describe $\mathbf{z}$.

B. Autoregressive Model based Feature Extraction

Autoregressive (AR) model of order m is defined as:

$$z(f) = \sum_{k=1}^m a_k z(f-k) + e_f(f) \quad (2)$$

where $a_k (k=1,2,\dots,m)$ are AR coefficients and $e_f(f)$ is forward prediction error. $a_k (k=1,2,\dots,m)$ are determined by the well-known Yule-Walker equation: $\mathbf{a} = \mathbf{R}^{-1}\mathbf{r}$.

From (2) we can see the information contained in HRRP is split into two parts: 1. the global frequency-evolving information represented by $\mathbf{a}$. This part reflects the target structure and can be utilized for recognition; 2. the reflectivity of HRRP return and local variations contained in $e_f(f)$. This part is noninformative for recognition and should be discarded.

Considering the amplitude of an HRRP is enlarged ρ times, the amplitude of its frequency spectrum becomes $\rho\mathbf{z}$ and thus $\mathbf{R}$ and $\mathbf{r}$ become $\rho^2\mathbf{R}$ and $\rho^2\mathbf{r}$. The new AR coefficients are obtained from $(\rho^2\mathbf{R})^{-1}(\rho^2\mathbf{r}) = \mathbf{R}^{-1}\mathbf{r}$, which remain unchanged. This indicates that the AR coefficients are invariant to HRRP amplitude scale. Thus, in total, the AR coefficients are initial-phase, translation and scale invariant.

Besides AR coefficients, the partial correlation (PARCOR) coefficients are also widely used in pattern recognition area. The PARCOR coefficient p_m of the m -th order AR model is defined as the cross-correlation coefficient between the forward and backward prediction error $e_f(f)$ and $e_b(f)$ of $(m-1)$ -th order AR model, i.e.:

$$p_m = \frac{E_f[e_f(f) * e_b(f-m)]}{\{E_f[e_f^2(f)]E_f[e_b^2(f-m)]\}^{1/2}} \quad (3)$$

with $e_f(f) = z(f) - \sum_{k=1}^{m-1} a_k z(f-k)$, $e_b(f) = z(f) - \sum_{k=1}^{m-1} a_k z(f+k)$

It has been proved that the PARCOR coefficient p_m is the same as the AR coefficients a_m for an m -th order AR model. Thus it has the same invariant properties as AR coefficients.

III. MIXTURE MODEL BASED FRAME PARTITION

Most papers divided the training data into frames at equal interval and modeled each frame by different statistical models. However, this method suffers from some limitations. First, since the measured HRRPs and further the features

extracted are non-uniformly distributed^[2], equal interval partition cannot ensure to allocate the similarly distributed samples into a same frame, which may result in the mismatch between samples and model. Second, frame number is dependent on the feature type and model employed. For equal interval partition, one needs to specify the frame number artificially, which is inappropriate. Here, we adopt a mixture model^[7] to deal with target-aspect sensitivity. The mixture model given in (4) considers all samples as a whole and can capture their global statistical characteristic.

$$p(\mathbf{u}) = \sum_{l=1}^L \alpha_l p(\mathbf{u} | \boldsymbol{\theta}_l), \text{ with } \sum_{l=1}^L \alpha_l = 1 \text{ and } \alpha_l > 0 \quad (4)$$

where $\mathbf{u}$ denotes feature sample, each component l corresponds to a target-frame, $p(\mathbf{u} | \boldsymbol{\theta}_l)$ denotes likelihood of the sample in the l -th frame and α_l is the prior probability of the l -th frame. The samples in each frame are assumed to follow Gaussian distribution, i.e., $p(\mathbf{u} | \boldsymbol{\theta}_l) = G(\mathbf{u} | \boldsymbol{\mu}_l, \boldsymbol{\Sigma}_l)$.

How to determine an appropriate frame number is a typical model selection problem. A Bayesian Ying Yang (BYY) harmony learning^[5] which implements model selection automatically during parameter learning is adopted here. The BYY learning is featured by the following harmony function:

$$H(p \| q) = \int p(\mathbf{R}, \mathbf{U}) \ln q(\mathbf{U}, \mathbf{R}) d\mathbf{U} d\mathbf{R}, \quad (5)$$

with $p(\mathbf{R}, \mathbf{U}) = p(\mathbf{R} | \mathbf{U})p(\mathbf{U})$ and $q(\mathbf{U}, \mathbf{R}) = q(\mathbf{U} | \mathbf{R})q(\mathbf{R})$.

where $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_N]$ represents the samples and $\mathbf{R}$ is its inner representation. Two types of Bayesian decomposition $p(\mathbf{R}, \mathbf{U}) = p(\mathbf{R} | \mathbf{U})p(\mathbf{U})$ and $q(\mathbf{U}, \mathbf{R}) = q(\mathbf{U} | \mathbf{R})q(\mathbf{R})$ are called Yang machine and Ying machine, respectively. Readers are referred to [9] for a systematic description of BYY system. Specifically for the mixture model, we have $\mathbf{R} = [\alpha_1, \alpha_2, \dots, \alpha_L]$ and (5) can be rewritten as:

$$H(p \| q) = \sum_{l=1}^L p(l | \mathbf{u}_n) \ln [\alpha_l q(\mathbf{u}_n | \boldsymbol{\theta}_l)], \quad p(l | \mathbf{u}_n) = \frac{\alpha_l q(\mathbf{u}_n | \boldsymbol{\theta}_l)}{\sum_{v=1}^L \alpha_v q(\mathbf{u}_n | \boldsymbol{\theta}_v)} \quad (6)$$

By maximizing $H(p \| q)$ in (6) with respect to each $\{\alpha_l, \boldsymbol{\mu}_l, \boldsymbol{\Sigma}_l\}$, the redundant components will be discarded. A gradient-based adaptive algorithm is sketched in the following:

Step 1: Choose a large enough L , set iteration index $\tau = 0$ and $H(\tau) = -\infty$, and initialize $\{\alpha_l, \boldsymbol{\mu}_l, \boldsymbol{\Sigma}_l\}$ randomly.

Step 2: Randomly pick a sample $\mathbf{u}_n$ from $\mathbf{U}$, and for each $l = 1, 2, \dots, L$, calculate:

$$p_{l,t} = p(l | \mathbf{u}_n) [1 + \ln p(l | \mathbf{u}_n) - \sum_{v=1}^L p(v | \mathbf{u}_n) \ln p(v | \mathbf{u}_n)]. \quad (7)$$

Step 3: Update parameters in a gradient manner:

$$\alpha_l^{new} = \frac{\exp(\beta_l^{new})}{\sum_{v=1}^L \exp(\beta_v^{new})}, \quad \beta_l^{new} = \beta_l + \eta(p_{l,t} - \alpha_l \sum_{v=1}^L p_{v,t}), \quad \beta_l = \ln \alpha_l; \quad (8)$$

⊙ if $\alpha_l \rightarrow 0$, discard component l , let $L^{new} = L - 1$ and continue;

$$\begin{aligned}\mu_l^{new} &= \mu_l + \eta p_{l,t}(\mathbf{u}_n - \mu_l); \\ \Sigma_l^{new} &= \Sigma_l + \eta p_{l,t}[(\mathbf{u}_n - \mu_l)(\mathbf{u}_n - \mu_l)^T - \Sigma_l]; \quad \tau^{new} = \tau + 1.\end{aligned}\quad (9)$$

Step 4: Compute $H(\tau^{new})$ by (6). If $H(\tau^{new}) - H(\tau) < \varepsilon H(\tau)$ holds, the algorithm converges; else, repeat step2 - step4.

Maximizing the harmony measure $H(p\|q)$ in Eq. (14), BYY learning will provide an intrinsic force to push $\alpha_l \rightarrow 0$ if the corresponding frame l is redundant and thus can be discarded. The extra frame removal step is labeled by “ $\odot$ ”.

IV. EXPERIMENTAL RESULTS

A. Data Description

The results presented in this paper are based on measured data. The center frequency and bandwidth of the radar are 5520MHz and 400MHz, respectively. The projections of plane trajectories onto ground plane are segmented as displayed in Fig. 1. There are 25600 HRRP samples in each segment except the 5th segment of Yark-42 which has 10240 HRRPs. The fifth and sixth parts of An-26, the sixth and seventh parts of Cessna and the second and fifth parts of Yak-42 are chosen as training data while the other parts are taken as test data. In addition, the dimensionality of HRRP is 256.

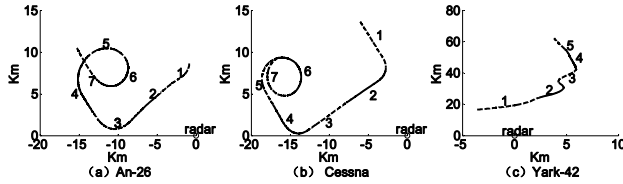


Figure 1. The projections of three target trajectories onto the ground plane.

B. Recognition Performance

For a fair comparison, all the experiments adopt the same equal interval partition as in [6] with each frame containing 1024 HRRP samples. Thus there are 50, 50 and 35 frames for An-26, Cessna and Yark-42, respectively. The Bayes classifier is employed for all recognition tasks. The average correct recognition rates (ACRR) based on AR/PARCOR coefficient feature versus model order are shown in Fig.2(b). As expected, the PARCOR coefficients slightly outperform the AR coefficients except the case of $m=1$. Best ACRRs of 93.6% and 95.1% are obtained for the AR and PARCOR features at model order $m=20$ and $m=22$, respectively. Afterwards, the ACRRs don't improve and even deteriorate somewhat. The reason is that some of the newly added coefficients are used to describe the noise or scintillation, i.e., the over-fitting problem.

In Figure 3, we give a comparison of ACRRs based on several typical schemes. It can be seen that AR and PARCOR features outperform the power spectrum of real HRRP^[6] by 3.6% and 5.1% while slightly inferior to real HRRP (modeled by Factor Analysis[1]) by 2.5% and 1%. The results indicate that the AR and PARCOR coefficients are just as viable as

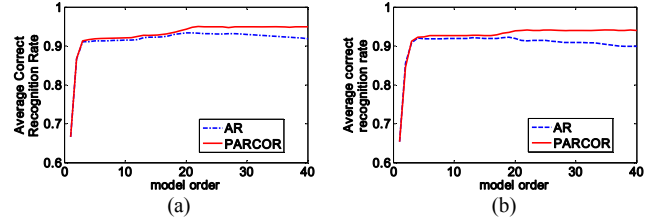


Figure 2. ACRRs versus model order based on different frame partition method: (a) equal interval partition; (b) mixture based partition.

other features for recognition. From table 1 we can see the AR and PCARCOR features possess the lowest dimensionality 22 and cost the least recognition time 4.6 ms, which greatly decrease the storage and computation burden of the recognition system.

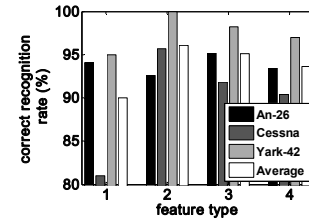


Figure 3. Correct recognition rates based on different features: (1). power spectrum of real HRRP; (2). real HRRP; (3). PARCOR; (4). AR.

TABLE I. COMPARISONS OF FEATURE DIMENSIONALITY AND TIME REQUIRED PER SAMPLE FOR DIFFERENT FEATURES

	Power spectrum	Real HRRP	AR/ PARCOR
Dimensionality	128	256	22
Test time required per sample (ms)	22	160	4.6

C. Frame Partition Results

Initialize the frame number of each target as 80 and set η and ε as 0.05 and 10^{-5} , respectively. It is very interesting to find that all the targets obtain the same frame number 1, irrespective of the model order. The ACRRs based on this partition results are shown in Fig. 2(b). As we can see, the ACRRs are almost the same as those in Fig. 2(a). These results indicate the AR/PARCOR feature is less sensitive to the variation of target aspect and the minor frame numbers can further decrease the recognition time.

D. The Effect of Sample Size

To characterize the recognition performance with respect to sample size, we uniformly divide the training samples of each target into 200 subsets and check the recognition accuracy against different fractions (1/200, 1/100, 1/50, 1/10, 1/5) of total training samples. The results are shown in Fig. 4(a), from which we can see the ACRRs based on AR and PARCOR features are robust to training sample size.

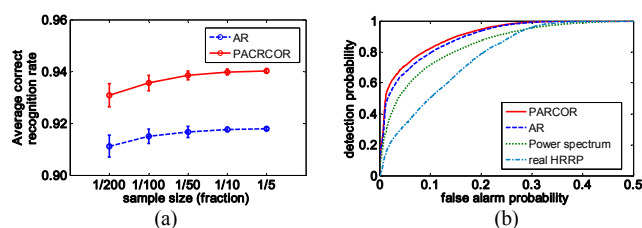


Figure 4. (a). ACRRs versus training data size. (b). ROC curves based on different features.

E. Rejection Performance

A crucial problem in radar HRRP recognition is how to discriminate in-class targets and confuser targets. When we cannot guarantee that all testing samples belong to the training set classes, rejecting those samples with a low degree of membership to these classes becomes important. We use 18000 HRRPs generated by XPATCH as the confuser targets to be rejected. Figure 4(b) comparatively shows the receiver operating characteristics (ROC) curves of aforementioned features. It is shown that the AR and PARCOR coefficients have the similar ROC curves and always achieve a higher detection probability than other two features for the same false alarm probability. This indicates that the AR and PARCOR coefficients are more effective in rejecting the confuser targets.

V. CONCLUSION

Two new features are proposed for HRRP based target recognition. A mixture model based frame partition method which is implemented by Bayesian Ying Yang harmony

learning is presented. Recognition results show the effectiveness and reliability of the proposed features.

VI. ACKNOWLEDGEMENT

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